

On Complexity, Computation, and Graph Homomorphisms

PhD Thesis Defence

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Outline

Overview

Borel Dichromatic Number and Σ_2^1 -Completeness

The L_o Dichotomy

Deep Computations: a Glimpse

Amoeba Graphs and the Feasible Edge-Replacement Group

Conclusion

Thesis section	Basis
§2.2 $\mathbb{L}_0$ dichotomy	Sole-authored; Fields Institute Set Theory Seminar (Mar. 2025)
§2.3 Dichromatic number	Sole-authored; Caltech Logic Seminar (Apr. 2026); paper in preparation
§2.4 Cantor game	Matos-Wiederhold-Salvetti, <i>Topology Proceedings</i> (2025)
§3 Deep Computations	Dueñez-Iovino-Matos-Wiederhold-Salvetti-Tall, arXiv (2026)
§4.2 Border tracing	P. Wiederhold-Matos-Wiederhold, <i>Theoretical Comp. Science</i> (2026)
§4.3 Recursive constructions	Eslava et al., <i>Aequationes Math.</i> (2025) & Caro et al., <i>Procedia CS</i> (2023)
§4.4 Amoeba trees	Gómez-González-Hansberg-MW, in preparation

What this thesis is about

“This thesis studies the cost of insisting on computable solutions.”

- ▶ Many existence proofs (Axiom of Choice, non-principal ultrafilters) give solutions with **no concrete description**.
- ▶ Imposing computability / definability on the solution is often still possible but costs something quantifiable.
- ▶ Recurring parallel throughout: **computer-science complexity** $\leftrightarrow$ **set-theoretic (descriptive) complexity**.
- ▶ Framed via **graph homomorphisms**: colourings, isomorphisms, and a group action on graphs are all homomorphism problems in disguise.

Three projects, one thread: the basis paradigm

A dichotomy of this shape recurs throughout:

either a problem has a simple, describable obstruction (a **basis**)
or it is provably too complex for one to exist.

Ch. 2a: Dichromatic number

Borel definability. Where does the countable-basis phenomenon ($\mathbb{G}_0$, $\mathbb{L}_0$) break down?

The $\mathbb{L}_0$ dichotomy (Ch. 2b, the one existing finite-threshold basis) gets a one-slide statement.

Ch. 3: Deep computations

Baire-class hierarchy of function spaces. When do limits of computation stay tame, rather than wild?

Ch. 4: Fer group

Polish group actions. When can a graph be turned into any labelled copy of itself, edge by edge?

Motivation: the basis landscape

Recall the $\mathbb{G}_0$ dichotomy (Kechris–Solecki–Todorćević, 1999): a single Borel graph $\mathbb{G}_0$ with

$$\chi_B(G) > \aleph_0 \iff \mathbb{G}_0 \rightarrow_B G.$$

Class	Countable basis?	Reference
$\chi_B(G) > \aleph_0$	Yes, size 1: $\mathbb{G}_0$	KST 1999
$\chi_B(G) > 2$	Yes, size 1: $\mathbb{L}_0$	CMSV; new proof here
$\chi_B(G) > k, k \geq 3$	No (Π_2^1 -complete)	Todorćević–Vidnyánszky
$\vec{\chi}_B(D) > \aleph_0$	No (size- $\aleph_0$ basis exists)	Raghavan–Xiao 2024
$\vec{\chi}_B(D) > k, k \geq 2$	?	

Setup

- ▶ G ranges over **analytic graphs**: $V(G) = X$ Polish, edge relation analytic in $X \times X$.
- ▶ $\chi_B(G)$: least k such that G admits a Borel proper k -colouring.
- ▶ A **directed graph** D has arcs (ordered pairs). A **dicolouring** $c: V(D) \rightarrow k$ requires each colour class to induce an *acyclic* subgraph.
- ▶ $\vec{\chi}(D) / \vec{\chi}_B(D)$: least k for a (Borel) dicolouring.
- ▶ Borel graphs/digraphs are studied via **nice codes** (Thornton): reals coding Borel sets, compatible with Σ_1^1/Π_1^1 definitions of the coded set.
“ Σ_2^1 -complete” is a statement about the complexity of the set of codes.

Main theorem

Proposition

The set of nice codes for Borel directed graphs admitting a Borel 2-dicolouring is Σ_2^1 -complete. Equivalently: the set of codes for Borel directed graphs with $\vec{\chi}_B(D) \geq 3$ is Π_2^1 -complete.

Corollary

There is no countable family of Borel directed graphs forming a basis for $\{D : \vec{\chi}_B(D) \geq 3\}$ under Borel homomorphism.

In fact, any basis at all must itself be Π_2^1 -hard as a set of codes.

Remark: symmetrization reduces to $k = 2$

For undirected G , let $\vec{G}$ replace each edge $\{u, v\}$ with both arcs $(u, v), (v, u)$.

$$\vec{\chi}_B(\vec{G}) = \chi_B(G),$$

since a vertex set is acyclic in $\vec{G}$ iff it is independent in G , and $G \mapsto \vec{G}$ is Δ_1^1 .

- ▶ So Σ_2^1 -completeness of Borel k -colouring **already gives** Σ_2^1 -completeness of Borel k -dicolouring for every $k \geq 3$.
- ▶ $k = 2$ is *not* covered; it is the new content.

Proof idea: the reduction

Lift the classical NP-completeness reduction (Bokal et al.) from 2-colouring of 3-uniform hypergraphs to 2-dicolouring of directed graphs.

- ▶ **Template gadget** F on $\{a, b, c, a', b', c'\}$: a colouring of the “shared” vertices $\{a, b, c\}$ extends to a 2-dicolouring of F iff it is **non-constant**.
- ▶ For hypergraph H : glue one gadget copy per hyperedge, identifying shared vertices with hypergraph vertices; keep auxiliary vertices private per copy.
- ▶ H is 2-colourable $\iff D(H)$ is 2-dicolourable.

Borel lifting (Thornton’s coding framework): every step of the construction is Δ_1^1 in the code of H .

Where this leaves the picture

Class	Countable basis?	
$\chi_B(G) > \aleph_0$	Yes, size 1: $\mathbb{G}_0$	
$\chi_B(G) > 2$	Yes, size 1: $\mathbb{L}_0$	this thesis
$\chi_B(G) > k, k \geq 3$	No (Π_2^1 -complete)	
$\vec{\chi}_B(D) > \aleph_0$	No (size- ϵ basis exists)	(in preparation)
$\vec{\chi}_B(D) > k, k \geq 2$	No (Π_2^1 -complete)	this thesis

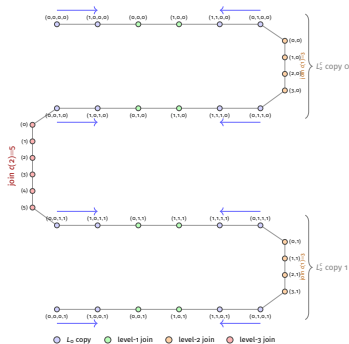
Every finite threshold, undirected and directed, is now settled.

The $\mathbb{L}_0$ dichotomy

Dichotomy (CMSV)

For any analytic graph G on a Polish space, $\chi_B(G) \geq 3 \iff \mathbb{L}_0 \rightarrow_c G$.

$\mathbb{L}_0$: an original proof by **condensation**,
in the spirit of Bernshteyn's proof of $\mathbb{G}_0$.



Why closures can misbehave

Quick reminder: a **CCS** $(L, \mathcal{P}, \Gamma)$ has states L , observable predicates $\mathcal{P}$, and a monoid $\Gamma \subseteq L^L$ of computations; **deep computations** are the closure $\bar{\Gamma}$.

The worry: closures of families of continuous functions need **not** stay “nice”.

The BFT dichotomy (Bourgain–Fremlin–Talagrand) fixes this

For pointwise-bounded $A \subseteq C_p(X)$: A ’s closure consists entirely of **Baire-1 functions (pointwise sequential limits)** $\iff$ A has **NIP** (no independence property).

NIP is exactly the dividing line between “the closure stays tame” and “the closure can go wild.”

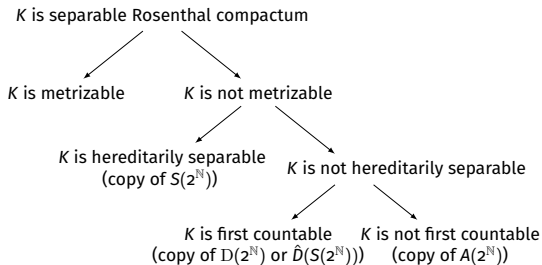
The Todorčević trichotomy

Theorem (Todorčević, 1999)

Let K be a separable Rosenthal compactum.

- ▶ hereditarily separable, non-metrizable $\Rightarrow$ copy of the **split Cantor set**;
- ▶ first countable, not hereditarily separable $\Rightarrow$ copy of an **Alexandroff duplicate**;
- ▶ not first countable $\Rightarrow$ copy of the **Alexandroff compactification** of a size- $\mathfrak{c}$ discrete set.

Our two examples (next slide) sit at opposite corners of this trichotomy.



Two examples: tame closures on Cantor space

Both live on the Cantor CSS $(2^{\mathbb{N}}, (\bullet_n)_n)$; both are countable, simple, *definable* predicates, yet their closures are richer. Both turn out tame (NIP), by BFT.

Threshold classifiers

$x \preceq_n w$: is $x \upharpoonright_n \preceq w$? Continuous, definable. **Closure** = “ultra-classifiers” $\preceq z$, $\prec z$ ($z \in 2^{\mathbb{N}}$): a *split Cantor set* $\cong 2^{\mathbb{N}} \times \{0, 1\}$, **hereditarily separable, non-metrizable**.

Prefix tests

$x \equiv_n w$: is w the length- n prefix of x ? Continuous, definable. **Closure** = **o**, finite-prefix tests, or point-*equality* tests $\bar{x} : y \mapsto \llbracket y = x \rrbracket$, separable, but **not first countable**.

Takeaway: “tame” doesn’t mean “simple,” just that every limit is describable.

The chapter's main theorem: BFT for CCS

Theorem (informal)

Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extendibility Axiom, with $\mathcal{P}$ and $\Delta \subseteq \Gamma$ countable. Then, on every shard, the deep computations generated by Δ form a Rosenthal compactum *if and only if* Δ has NIP, feature by feature.

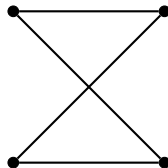
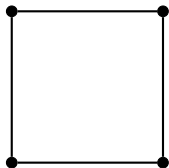
When this holds, every deep computation is a **sequential** pointwise limit of ordinary computations.

Combined with the Todorčević trichotomy, this yields a hierarchy of **four** tameness classes, $\text{NIP}_3 \Rightarrow \text{NIP}_2 \Rightarrow \text{NIP}_1 \Rightarrow \text{NIP}$.

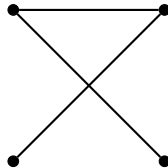
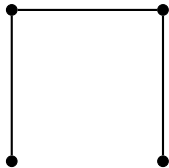
Our two examples above witness two of these separations; the thesis exhibits examples separating every level.

The Fer Group (Caro–Hansberg–Montejano)

C_4 **is rigid**: two isomorphic labelled 4-cycles, connected by *no* sequence of feasible edge replacements.



The puzzle: two isomorphic labelled copies of P_4 . Can one reach the other?



Setup: the Fer group

Fix a labelling $\lambda: V(G) \rightarrow X$. For $e \in E(G)$, $f \in E(G^c)$ non-edge, write $\sigma :: (e \rightarrow f)$ if the label permutation σ induces an isomorphism $\varphi_\sigma: G \rightarrow G - e + f$.

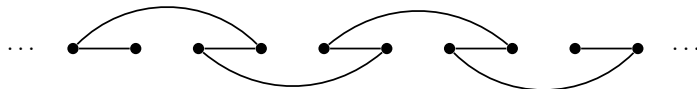
- ▶ $\text{Fer}(G) := \langle \sigma : \sigma :: (e \rightarrow f) \text{ for some } e, f \rangle \leq S_X$, the **canonical generators** include $A(G) := \{\sigma : \varphi_\sigma \in \text{Aut}(G)\}$ and all single-edge moves.
- ▶ $1 \leq A(G) \leq \text{Fer}(G) \leq S_X$.
- ▶ G is a **local amoeba** if $\overline{\text{Fer}(G)} = S_X$ (every relabelling is reachable); **rigid** if $\overline{\text{Fer}(G)} = A(G)$ (no non-automorphism move exists).
- ▶ G is a **global amoeba** if $G \cup tK_1$ (add t isolated vertices) is a local amoeba for some $t \geq 1$.

Example: $\text{Fer}(C_4) = A(C_4) = D_4$ (rigid); $\text{Fer}(P_4) = S_4$ (local amoeba).

A twist at infinity: the bi-infinite path $P_{\mathbb{Z}}$



Bi-infinite path



Bi-infinite twisted path

$\text{Fer}(P_{\mathbb{Z}}) = A(P_{\mathbb{Z}})$; it is **rigid**, just like C_4 . (The one-directional path P_{ω} , by contrast, is a local amoeba.) This entire infinitary strand is **sole-authored** (the existing Fer -group literature only treats finite graphs).

Classification of global amoebas (with Caro–Hansberg–Montejano)

For non-empty G : G is a global amoeba $\iff G \cup K_1$ is a local amoeba $\iff$ every $\text{Fer}(G)$ -orbit contains a degree-1 vertex.

Fibonacci-type trees T_k (with Eslava et al.)

T_k (built by recursively joining max-degree vertices, $|V(T_k)| = 2F_k$) is a local amoeba for every k , with balancing number Θ -bounded in terms of Fibonacci numbers.

The infinitary case: Baire-category constant balancing

Extending to X countably infinite: S_X becomes a Polish group, $\text{Fer}(G)$ an analytic subgroup.

Theorem (infinitary analogue of the Caro et al. characterization)

Let $c: [X]^2 \rightarrow 2$ be a Baire-measurable colouring of a perfect Polish space X with non-meagre colour classes. Then some colour i , vertex v_0 , and Cantor set C satisfy: every edge v_0-C has colour i , and $[C]^2$ is monochromatic in colour $1 - i$ (a continuum-size star *and* a continuum-size clique).

Theorem (infinitary analogue of the Eslava et al. identity)

Let X be a perfect Polish space, G a finite global amoeba, and $c: [X]^2 \rightarrow 2$ a Baire-measurable colouring with non-meagre colour classes. Then K_X , coloured by c , contains a **balanced** copy of G .

Same localization + Cantor-extraction argument as the previous theorem, then a **discrete intermediate value** argument: walk a chain of feasible edge replacements from a red-heavy copy of G to a blue-heavy one, and some copy along the way must be balanced.

Applied side: complexity & border tracing

- ▶ $O(n \log n)$ **decomposition.** Factoring an arbitrary permutation of T_k into feasible edge replacements: the published algorithm's analysis gives $\Theta(n^2)$ time (Eslava et al., Thm. 11).
- ▶ **Computational framework.** The `fer` object formalism is implemented as an open-source object-oriented framework for experimenting with feasible edge replacements.
- ▶ **Border tracing** (peripheral side project): a corrected version of Voss' Border Mesh Determination Algorithm for oriented adjacency graphs of polygonal tilings (square/triangular/hexagonal).

Open direction: genuinely rigid graphs

Schweitzer–Schweitzer (2017, resolving a Nešetřil conjecture): exactly 18 minimal asymmetric graphs. Does an analogous finite classification hold for Fer-rigidity?

- ▶ Naive translation fails: odd cycles C_{2n+1} are rigid *vacuously*.
- ▶ Fix: G is **genuinely rigid** if $\text{Fer}(G) = A(G)$ *and* some degree-preserving replacement exists.
- ▶ Smallest example: $K_{2,3}$.

Question

Are there finitely many *minimal* genuinely rigid graphs?

Summary

“This thesis studies the cost of insisting on computable solutions.”

- ▶ **Dichromatic number:** the basis paradigm for finite thresholds is now fully settled, undirected *and* directed. The last piece was Σ_2^1 -completeness of Borel 2-dicolouring.
- ▶ $\mathbb{L}_O$ **dichotomy:** a new condensation-style proof, adapting the KST/Bernshteyn argument from $\mathbb{G}_O$ to a setting with a dynamically-changing parameter.
- ▶ Fer **group:** a Polish group action encoding a purely combinatorial notion of rigidity, now with a classification of global amoebas, an infinitary (Baire-category) extension, and a genuinely improved algorithm.

Three different DST languages: Borel definability, condensation/reflection, Polish group actions, applied to the same underlying question: what happens to a graph-homomorphism problem once you demand a definable witness?

Open problems

- ▶ **Dichromatic number:** does every basis for $\vec{\chi}_B(D) > \aleph_0$ (Raghavan–Xiao) have size $\mathfrak{c}$, or is there a smaller one? Does a countable basis exist for structurally simpler classes (e.g. directed graphs generated by a single Borel function)?
My conjecture: there is no countable basis. Preparing draft for submission.
- ▶ **Fer group (infinitary):** characterize the countably infinite, locally finite trees that are local amoebas. Is $\overline{\text{Fer}(T)}$ always amenable for non-rigid trees? Is the local-amoeba property itself Borel?
- ▶ **Fer group (finite):** are there finitely many minimal genuinely rigid graphs?

Thank you

Questions?

Backup: dichromatic number — minimal monochromatic cycles

Why does a 2-dicolouring of H 's gadget copies glue to a 2-dicolouring of $D(H)$? The naive hope “each gadget is locally fine, so the whole graph is fine” needs repair, since a monochromatic cycle need **not** stay inside one gadget copy.

1. Take a monochromatic directed cycle C of *minimal length* (for contradiction).
2. Arcs between original hypergraph vertices respect a fixed linear order $\Rightarrow C$ cannot lie entirely in V .
3. C cannot lie entirely in one gadget F_e either (each $\sigma(e)$ is a genuine 2-dicolouring of F_e).
4. So C crosses a gadget via a segment $x \rightarrow \dots \rightarrow y \rightarrow \dots \rightarrow z$, y auxiliary, $x, z \in \{a, b, c\}$.
5. Minimality $\Rightarrow$ no shortcut arc $x \rightarrow z \Rightarrow (x, z) \notin E(F) \Rightarrow$ (checking F 's three shared-to-shared arcs $a \rightarrow b, a \rightarrow c, b \rightarrow c$) forces the arc $z \rightarrow x$.
6. But then $z \rightarrow x$ plus the segment $x \rightarrow \dots \rightarrow z$ is a *monochromatic cycle entirely inside F_e* : a contradiction.

Backup: $\mathbb{L}_0$, two ways to prove a perfect set theorem

Both strategies analyze the pruned tree $T \subseteq 2^{<\omega}$ with $C = [T]$:

- ▶ **Terminal approximation:** iteratively delete nodes with ≤ 1 branch above them, through the countable ordinals. (Original KST proof of the $\mathbb{G}_0$ dichotomy.)
- ▶ **Condensation:** keep only nodes above which C is uncountable, in *one step*. (Bernshteyn's newer proof of $\mathbb{G}_0$.)

Equivalent in strength for $2^{<\omega}$, but condensation is more economical and, unlike terminal approximation, needs a form of choice that fails under AD.

This chapter gives a new proof of the $\mathbb{L}_0$ dichotomy (Carroy–Miller–Schrittesser–Vidnyánszky, originally proved by terminal approximation) via condensation, in direct analogy with Bernshteyn's argument for $\mathbb{G}_0$, with the added twist of a **dynamically-changing parameter**.

Backup: $\mathbb{L}_0$ setup

- ▶ Fix $c \in \omega^\omega$. Build finite paths L_n^c recursively: L_{n+1}^c joins *two copies* of L_n^c by a fresh path of $c(n) + 2$ edges.
- ▶ $\mathbb{L}_c$ is the natural limit graph: vertices are (m, k, x) with $x \in 2^\omega$ recording, at each stage, which of the two copies a point falls in.
- ▶ If c is unbounded and odd-valued: $\mathbb{L}_c$ is an acyclic graph of degree ≤ 2 , and $\chi_B(\mathbb{L}_c) = 3$.
- ▶ Fixing one canonical such c_0 : $\mathbb{L}_0 := \mathbb{L}_{c_0}$.

Backup: $\mathbb{L}_0$ main theorem

The dichotomy follows from combining two theorems:

Theorem A

For any analytic graph G : either $\chi_B(G) \leq 2$, or $\mathbb{L}_c \rightarrow_c G$ for some unbounded, odd-valued c .

Theorem B

For c, d unbounded and odd-valued, $\mathbb{L}_c$ and $\mathbb{L}_d$ are continuously homomorphically equivalent.

Theorem B is purely combinatorial and does not depend on how Theorem A is proved. **Theorem A is the new condensation-style argument.**

Backup: $\mathbb{L}_0$ proof strategy: condensation

Central device: a σ -ideal of **small** sets of homomorphisms $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$.

- ▶ $\Phi(A, k)$: A is analytic and all odd walks of G with endpoints in A have length $\leq 2k - 1$.
- ▶ $\mathcal{L}$ is *tiny* if $\Phi(\mathcal{L}(u), k)$ for some vertex u and some k ; *small* = countable union of tiny sets; *large* = not small.
- ▶ Key fact: if $\Phi(A, n)$, then A 's connected component is contained in an invariant Borel 2-colourable set (via analytic $\Rightarrow$ Borel separation).

So: $\chi_B(G) > 2 \Rightarrow X$ itself is large (as $\text{Hom}(\bullet, G)$). Recursively build a nested sequence of *large* Borel sets $\mathcal{L}_n \subseteq \text{Hom}(L_n^c, G)$ of shrinking diameter, dynamically choosing $c(n)$ at each stage; a continuous homomorphism $\mathbb{L}_c \rightarrow G$ falls out as the Cantor-intersection limit.

Backup: $\mathbb{L}_0$ key lemma: largeness survives doubling

The crux is showing largeness can always be preserved when doubling $L_n^C \rightarrow L_{n+1}^C$:

Lemma

If $\mathcal{L} \subseteq \text{Hom}(L_n^C, G)$ is large, then for any N there is an odd $d(n) \geq N$ such that $\mathcal{L}^{+d}$ (the set of homomorphisms restricting to $\mathcal{L}$ on both new copies) is again large.

Why: suppose not, then every choice of gap length decomposes $\mathcal{L}^{+d}$ into countably many tiny pieces. Each tiny witness (an analytic set with a Φ -bound) can be reflected up to a **Borel** set with the same bound, via the **First Reflection Theorem** (Π_1^1 -on- Σ_1^1). Removing all these countably many Borel-tiny pieces from $\mathcal{L}$ leaves a large remainder, which produces a genuine counterexample, a contradiction.

This single reflection step replaces the transfinite construction used in the original $\mathbb{L}_0$ proof.

Backup: $\mathbb{L}_o$ — where this stands

- ▶ New proof, structurally parallel to Bernshteyn's condensation proof of the $\mathbb{G}_o$ dichotomy — strengthens the analogy between the two dichotomies.
- ▶ Main new ingredient: replacing the independence-based σ -ideal (used for $\mathbb{G}_o$) with the odd-walk-bounded Φ -ideal, needed because $\mathbb{L}_c$ requires a dynamically-changing parameter.
- ▶ First disseminated at the Fields Institute Set Theory Seminar (March 28, 2025).
- ▶ Open: do analogous condensation-style proofs exist for other Borel-combinatorics dichotomies?

Backup: contribution — §2.4 Cantor game

MW, Salvetti (*Uncountable sets and an infinite linear order game*, Topology Proceedings 2025).

My contribution: equal contribution throughout — the results, the writing, and the submission.

Dueñez, Iovino, MW, Salvetti, Tall (*Complexity of Deep Computations via Topology of Function Spaces*, arXiv 2026).

My contribution: constructed the two examples used to separate the NIP_i hierarchy — threshold classifiers on Cantor space (witnessing $\text{NIP}_2 \not\equiv \text{NIP}_3$) and prefix/equality tests (witnessing $\text{NIP} \not\equiv \text{NIP}_1$). Presented this material extensively at the Fields Institute seminar series (2024–2026) and contributed substantially to drafting and revising the paper.

P. Wiederhold, MW (*Border tracing in oriented adjacency graphs...*, TCS 2026).

My contribution: co-developed the border-tracing algorithms, including the corrected End Condition Test, and their edge-case analysis; independently designed, implemented, and documented an object-oriented software framework realizing the algorithms, with a test suite validating contour detection on square, triangular, and hexagonal tilings.

Eslava, Hansberg, MW, Ventura (*New recursive constructions of amoebas...*, Aequationes Math. 2025) — Fibonacci-type trees T_k , balancing number bounds.

My contribution: substantial, co-equal contribution to the main result that the trees are local amoebas; equal contribution to the remaining joint sections. Independently wrote Section 5 (the recursive algorithm and its complexity analysis) and independently designed and implemented the accompanying object-oriented software framework, including the abstract classes for feasible edge replacements.

Backup: contribution — §4.3 Constant balancing number

Caro, González, Hansberg, Jácome, MW, Montejano (*Graphs with constant balancing number*, Procedia CS 2023).

My contribution: equal contribution to the paper's main results, the structural characterization of constant-balancing-number graphs.

Gómez, González, Hansberg, MW (*Amoeba trees*, in preparation) — structural theory of $\text{Fer}(T)$ vs. $\text{Aut}(T)$.

My contribution: originated the paper's central conjecture — that the Fer group of every global amoeba tree has at most two orbits — and extended the object-oriented framework with a statistics database used to find counterexamples and formulate further conjectures; equal contribution to the proofs and main results.

Backup: sole-authored work

Sole-authored, under Spencer Unger's supervision:

- ▶ §2.2 — the new $\mathbb{L}_0$ proof;
- ▶ §2.3 — the Σ_2^1 -completeness / dichromatic number result;

Sole-authored:

- ▶ §4.5 — the entire infinitary Fer -group strand.

Backup: in what sense do these structures “compute”?

A **Compositional Computation Structure** $(L, \mathcal{P}, \Gamma)$: L = states, $\mathcal{P}$ = observable real-valued features, $\Gamma \subseteq L^L$ a **monoid of literal state-transformation maps** ($\text{id} \in \Gamma$, composition = running one computation after another).

- ▶ Elements of Γ are the **computations**: they “compute” in the ordinary sense, each a function $L \rightarrow L$, closed under composition.
- ▶ **Deep computations** $\xi \in \bar{\Gamma}$ (closure in $\mathcal{L}^L$) are possibly-infinitary *limits* of literal computations — not composable in general.
- ▶ γ is **extendable** if it has a K -extension: continuous on each compact type-shard $\mathcal{L}[r_\bullet]$ (shards = bounded-magnitude, bounded-precision pieces exhausting L).
- ▶ **Extendibility Axiom**: every $\gamma \in \Gamma$ is extendable.

So: standard computations compute by definition; deep computations compute in the extended sense of being continuous limits, uniquely determined on every bounded shard.

Backup: limitations & what I'd do differently

- ▶ Dichromatic number: the Raghavan–Xiao basis for the uncountable directed threshold is size $\aleph_1$ — unknown whether that's optimal, or whether restricting to structurally simpler digraph classes helps.
- ▶ For group: the infinitary theory currently only has trees; characterizing local amoebas among general countably infinite graphs is open, as is the Borel complexity of the local-amoeba property itself.
- ▶ Genuinely rigid graphs: only checked computationally up to 7 vertices; no structural (forbidden-subgraph-style) proof strategy yet in view.
- ▶ ...